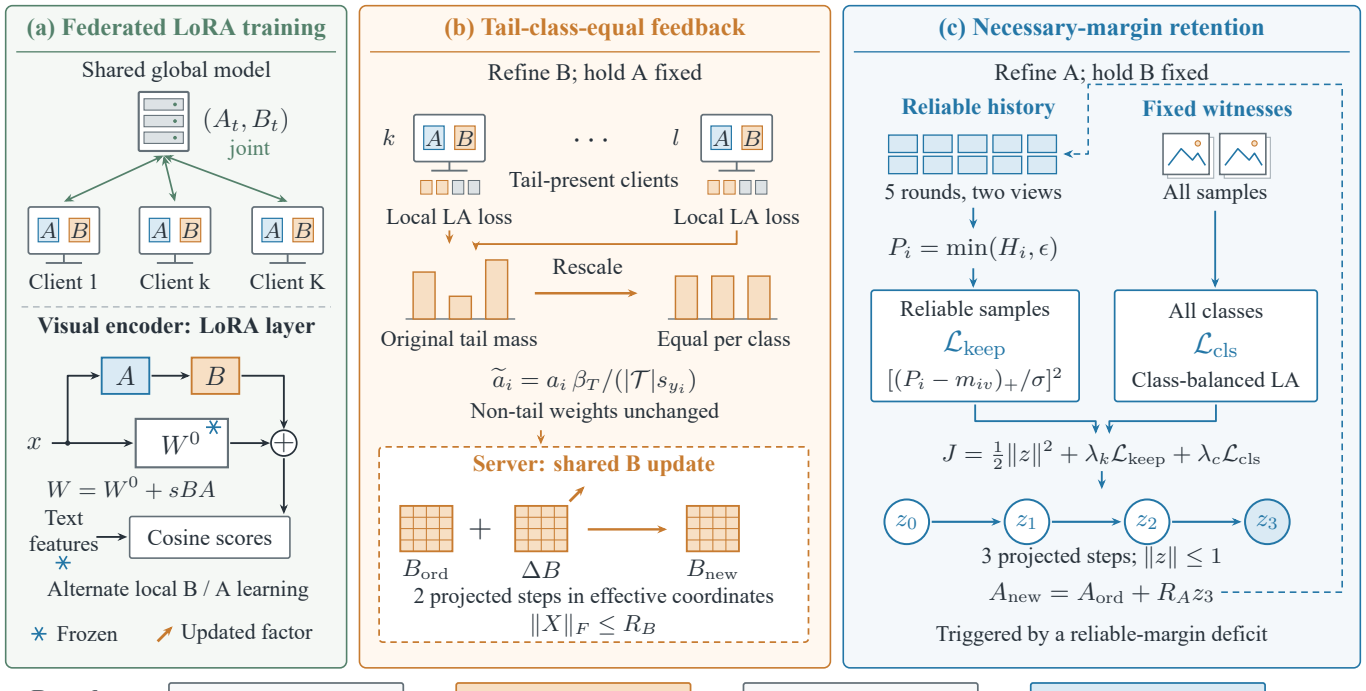

(a) Federated LoRA training
Shared global model
(A_t, B_t)
joint
A B
Client 1
Client k
Client K
Visual encoder: LoRA layer
A
B
x
W^0
$W = W^0 + sBA$
Text features
Cosine scores
Alternate local B / A learning
Frozen
Updated factor
(b) Tail-class-equal feedback
Refine B; hold A fixed
k
l
Tail-present clients
Local LA loss
Local LA loss
Rescale
Original tail mass
Equal per class
$\widetilde{a}_i = a_i\, \beta_T/(|\mathcal{T}|s_{y_i})$
Non-tail weights unchanged
Server: shared B update
B_{ord}
ΔB
B_{new}
2 projected steps in effective coordinates
$\|X\|_F \le R_B$
(c) Necessary-margin retention
Refine A; hold B fixed
Reliable history
Fixed witnesses
5 rounds, two views
All samples
$P_i = \min(H_i, \epsilon)$
Reliable samples
$\mathcal{L}_{\text{keep}}$
$[(P_i - m_{iv})_+/\sigma]^2$
All classes
$\mathcal{L}_{\text{cls}}$
Class-balanced LA
$J = \frac{1}{2}\|z\|^2 + \lambda_k \mathcal{L}_{\text{keep}} + \lambda_c \mathcal{L}_{\text{cls}}$
z_0
z_1
z_2
z_3
3 projected steps; $\|z\| \le 1$
$A_{\text{new}} = A_{\text{ord}} + R_A z_3$
Triggered by a reliable-margin deficit
Round t
Local B + joint
B feedback (b)
Local A + joint
A retention (c)
(A, B)
Every round
$t = 30, 40, \ldots, 100$
$t \le 90$
$t \le 90$